MERRA21C-ML: GENERATIVE CONUS DOWNSCALING

Native GEOS-FP (~25 km) → variable-specific conditional flow matching → six fields on the HWT (~3 km) grid

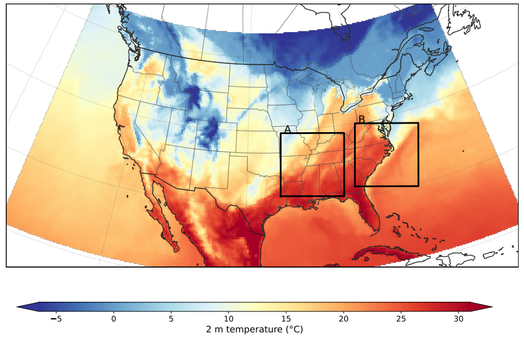
GEOS-FP + high-resolution GEOS replay + v4.1 predictions | 2026-03-07 19:30:00 UTC | epoch 243, EMA, 8 members, spectral

A COARSE WEATHER INPUTS

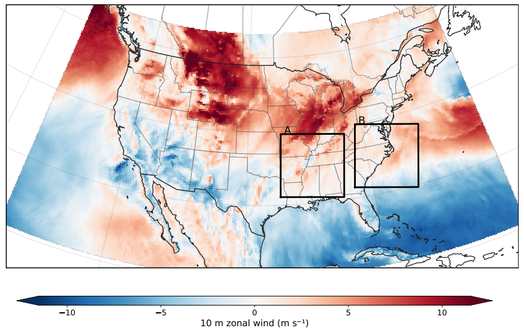
Current + previous hour; static and time features

Native 0.25° × 0.3125° cells

2 m temperature



10 m wind: zonal component



Six weather fields at t − 1 h and t

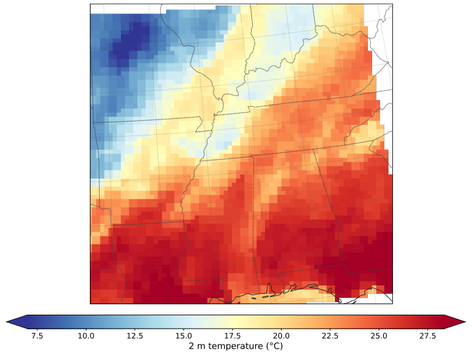
Local 192² patches + broad 576² context

Terrain, land/sea, coordinates, cell area, time

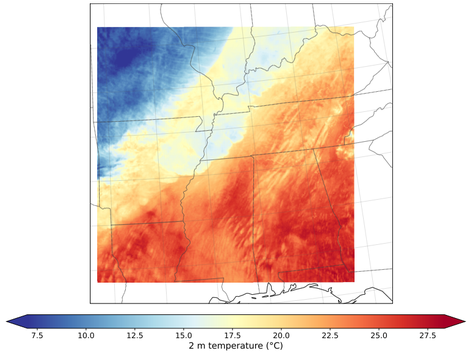
B SAME REGION, TWO GRIDS

Identical extent and color scale; no display interpolation

~25 km: native cells



~3 km: GEOS replay reference (HWT)



Temporal pairing and training-only statistics

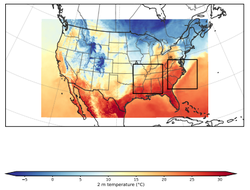
States: midpoint | Rain: ¼ P_{-30} + ½ P_0 + ¼ P_{+30}

Chronological splits with gaps; packed training data

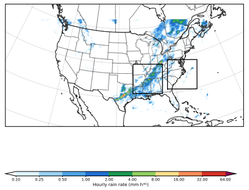
C SIX JOINT OUTPUT FIELDS

Actual v4.1 model member on the high-resolution GEOS grid

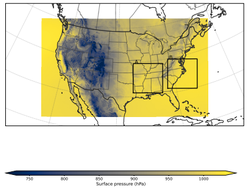
2 m temperature



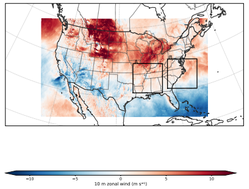
Hourly mean rain rate



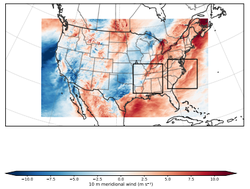
Surface pressure



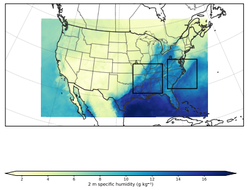
10 m zonal wind



10 m meridional wind



2 m specific humidity



Variable-specific generative coordinates

Rain: direct $\sqrt{(1 + P/P_0)} - 1$; P_0 = 1 mm h⁻¹

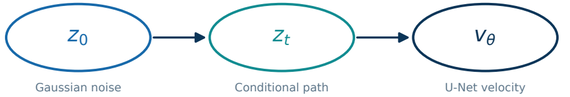
T, pressure, winds: regression residuals

Humidity: coarse-field residual

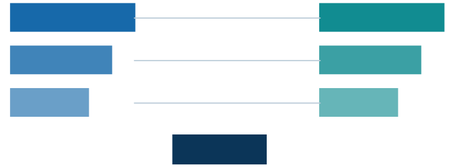
D REGRESSION → GENERATION

Frozen conditioner + local and broad-context U-Net

CONDITIONAL FLOW MATCHING



$z_t = (1 - t)z_0 + tz_1$ | target: $z_1 - z_0$



Residual blocks + broad-context conditioning

AREA- AND CHANNEL-WEIGHTED VELOCITY LOSS



192² input = 128² core + 32-pixel halo

Loss supervises core and halo

SYNCHRONIZED TILED ENSEMBLE SAMPLING

Blend tile velocities → shared-state Heun update

Independent seeds → multiple conditional members



One coarse input → 8 generated high-resolution members

Raw / spectral: paired members, no retraining

Maps: native GEOS-FP, HWT dynamical GEOS replay reference and actual v4.1 predictions | 2026-03-07 19:30:00 UTC | 64 Heun steps, spectral.

Temperature zooms share extent and scale. Original map panels are preserved; flow matching performs learned downscaling to the GEOS reference grid.